Proposal: Investigation on Optimal Shoe Sole Design

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***Abstract*— 250 words**

I. Introduction

Numerous designs of sports shoes on the market feature thick and flexible soles. Although the designs aim to maximize user comfort, little has been investigated on how they improve the quantitative metrics for the user, such as metabolic cost and ground impact force.

It is also worth investigating on how the choice of material and design of the soles can be optimized for people from different demographics. Additionally, variation in environments, such as the change in surface stiffness (cement vs asphalt) and ground inclines, can affect the performance of soles.

Motivated by the above questions, our team decides to investigate on how sole stiffness, and other potential design parameters such as thickness, can be optimized for different demographics and environments.

We choose the task of running, a physically demanding task, which can lead to injury especially to the knee joint under unfavorable conditions. Despite the complexity of human locomotion, we plan to model running based on the provided hopper model, due to their similarities in dynamics.

II. Background

[1] measures the bending stiffness of shoes, as the human metatarsophalangeal (forefoot/toe) joint exerts a bending moment during running. They conclude that the human forefoot stiffness is much higher than the shoe stiffness, and dominates the combined stiffness of the foot and shoe, hence shoe bending stiffness might not have a significant effect on running performances; however, human foot stiffness decreases near toe-off, hence shoe stiffness might have a larger effect then. [2] finds that adding flexibility to the prosthetic toe joint is able to reduce push-off work when walking.

[3] creates and compares several models of Ground Reaction Forces (GRF) that take the human biomechanics and shoe soles into consideration. They discovered that the GRF profile of the hard sole process a sharper impact peak than that of the soft sole, while other parts of the profiles are similar, as in Fig. 2. They create several models of the GRF, either with linear springs and dampers or nonlinearly, but both models give similar results.

III. Methods

A. Model Overview

We are planning to build our simulation based on the provided `Fullhopper.slx` model, assuming that the performance in hopping can approximate the performance in running to some degree.

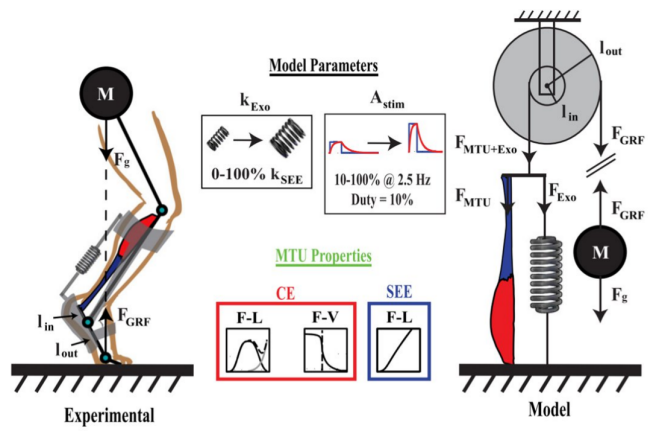

Fig. 1: The hopper model which our simulation is based on.

Unlike in walking, humans when running cannot utilize the exchange between potential and kinetic energy. Instead, humans are storing and returning mechanical energy through elastic tissues, and the CoM motion resembles bouncing. In addition, there are periods of aerial phases when running where no foot is contacting the ground [4]. These characteristics are also present in the `Fullhopper.slx` model, making it a simplified and tractable approximation to the running dynamics.

However, considering that only one foot is impacting the ground when running, while two feet are impacting the ground when hopping, explaining the load mass in the model being 35 kg. We may need to double the load mass in the provided model for a more realistic approximation.

To model the effect of sole, we utilize the fact that the compression of the sole contributes to the body COM displacement; only CoM displacement after subtracting this compression - the displacement with respect to the bottom of foot, but not the ground, will influence MTU kinematics.

B. Model Formulation

The main modification to the original model is to add a block into the `LoadDynamics` block, which takes GRF as input and outputs the displacement of the sole dp_{sole}. This displacement will affect the kinematics by being subtracted from the body CoM displacement dp_{COM}, before the latter is being multiplied by $-EMA$ to obtain MTU length change. We denote the CoM displacement after this subtraction by

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dp_{sub}. Then, the relevant dynamics will become:

$$\begin{aligned} dp_{sole} &= x_k(F_{GRF}) \\ dp_{sub} &= dp_{COM} - dp_{sole} \\ dL_{MTU} &= -EMA \times dp_{sub} \end{aligned}$$

Where $x_k(F)$ is a function that outputs the sole displacement from the force F on the sole.

Note that in this formulation, dp_{sole} is negative, as the sign of displacement should be the same for both dp_{COM} and dp_{sole}. We follow the convention that dp_{COM} is positive when moving upwards, and the sole can only be compressed (surface moving downwards).

1) Mass-Spring-Damper Model: The acceleration of the sole CoM can be approximated as half of the acceleration of the upper sole surface (?).

We denote the reaction force from the ground acting on the sole as F_{RS}, and the reaction force from the sole acting on the human as F_{RH}. The dynamics of the entire sole is:

$$m(\frac{1}{2}\ddot{x}) = F_{RS} - F_{RH}$$

The dynamics of the upper surface is:

$$m(\frac{1}{2}\ddot{x}) = -F_{RH} - kx - b\dot{x}$$

We need to obtain either F_{RS} or F_{RH} to solve for x; however, the only relevant force from the original model is the ground reaction force, $F_{GRF} = F_{MTU}EMA$ when the human is in stance. We can only assume $F_{RH} = F_{GRF}$ - the reaction force equals to the GRF acting on CoM, even though this is unrealistic since the contact surface is not rigid, the reaction force on human F_{RH} cannot immediately respond to the MTU force. The system then becomes:

$$\begin{aligned} m(\frac{1}{2}\ddot{x}) &= -F_{GRF} - kx - b\dot{x} \\ \ddot{x} &= \frac{2}{m}(-F_{GRF} - kx - b\dot{x}) \\ s.t.\ x(t_0) &= \frac{-Mg}{k(x(t_0))} \end{aligned}$$

This formulation is achievable in Simulink utilizing integration blocks. Note that, instead of subtracting CoM displacement by x, we need to subtract acceleration of CoM by $\ddot{x}$; this leads to the position of CoM subtracted by x, as they are related by integration.

However, it requires either $k(x)$ to be well-defined for all possible x, or both functions $F(x)$ (for $\dot{x}$) and $x(F)$ (for $x(0)$) exist and are inverses to each other - the stress-strain curve has to be bijective or strictly monotonic.

Try with entire human - sole system:

$$\begin{aligned} (M+m)\ddot{x}_{COM} &= \sum F \\ M\ddot{x}_h + m(\frac{1}{2}\ddot{x}) &= F_{RS} - (M+m)g \\ \ddot{x}_h + \ddot{x} &= \ddot{x}_{COM} \end{aligned}$$

Still unrelated with MTU force, model is degenerate...

2) Spring-Damper Model: This model is formulated under the assumption that, at each instant, the sole is under force equilibrium - its reaction force equals to the GRF acting on CoM, implying that the upper surface of the sole has net force close to zero, $-kx - b\dot{x} - F_{GRF} = 0$ - the acceleration is negligible compared to the human kinematics. We formulate the ODE by rearranging terms:

$$\begin{aligned} \dot{x} &= \frac{1}{b}\left((-F_{GRF}) - k(x)x\right) \\ s.t.\ x(t_0) &= \frac{-Mg}{k(x(t_0))} \end{aligned}$$

As in Mass-Spring-Damper model, this formulation is achievable in MATLAB and we need to subtract the velocity of CoM by $\dot{x}$, it also poses constraints on the formulation of stress-strain curve.

3) Spring Model: If we assume that the system is always in steady state, $\dot{x} \approx 0$ and we can drop the damping term. The displacement can therefore be calculated as $x = \frac{F}{k(x)}$ or $x(F)$, hence we do not need to formulate $k(x)$ or $F(x)$ explicitly, there are less constraints on $x(F)$ than in the case with damping.

C. Simulation Modification Details

When displacement is modified, the following will also be affected: 1. the stance condition, which is $dL_{MTU} > 0$ (MTU lengthened) in the original model, equivalent to $dp_{COM} < 0$ (COM lower than initial height); 2. the initial condition of displacement, which is $dp_{COM}(t_0) = 0$ in the original model.

We need to first define a coordinate of displacement, namely its origin with zero displacement.

Since $dp_{sub} = dp_{COM} - dp_{sole}$, and when static, we assume MTU length to be unchanged,

$$\begin{aligned} dL_{MTU} &= -EMA \times dp_{sub} = 0 \\ \therefore dp_{sub} &= 0 = dp_{COM} - dp_{sole} \\ dp_{COM} &= dp_{sole} \end{aligned}$$

When static, the force on sole is Mg,

$$\begin{aligned} \therefore dp_{sole}(t_0) &= x_k(-Mg) \\ dp_{COM}(t_0) &= dp_{sole}(t_0) = x_k(-Mg) \end{aligned}$$

which is the initial condition of dp_{COM}. The origin of dp_{COM} is where $dp_{sole} = 0$ - where the human is standing on an uncompressed sole.

As for stance condition, due to the sole force equilibrium formulation, the human will receive a reaction force as soon as the uncompressed sole touches the ground, that is $dp_{COM} = 0$. The stance condition is therefore still $dp_{COM} < 0$, but with a shifted origin defined above.

IV. Experiments

A. Variables to Examine

We provide a list of independent variables of interest for this project. We may decide on a subset of them, depending on the available time and workload for the class.

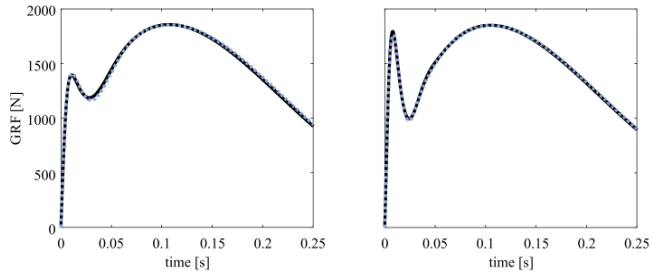

Fig. 2: GRF profiles of running with soft (left) and hard (right) soles [3].

1) Demographic Differences: Differences in population, such as gender and age, can affect locomotion performance and optimal insole design.

Relevant quantitative parameters can be weight, MTU length, maximum muscle force, tendon stiffness, activation magnitude and frequency.

2) Sole Properties: The basic deliverable is to alter the value of spring constant for the sole modeled as a linear spring. For more realistic results, we might model different insole material stress-strain curves. We can also investigate the effect of sole thickness.

B. Environment and Task Differences

Stiffness and incline of the surface can be considered and modeled as explained in III-C. In addition, terrain or task differences can lead to changes in gait/jump height and frequencies, changing the required activation.

V. Results

A. Hypotheses of Outcomes

From the GRF plots obtained in [3], shown in Fig. 2, we can see that the GRF profile when running with a hard sole has a much higher force peak at the beginning due to the foot compacting, compared to that of the soft sole. We can infer that this higher impact force results in more energy loss, leading to the hypothesis that as the sole stiffness increases, the user has to spend more metabolic cost to make up for this loss to run at the same speed.

Those with degraded tendon performances, such as the elderly, may require a lower spring stiffness to compensate for reduced energy storage and return abilities. However, our model does not take stability and other important aspects into consideration, hence this result may not transfer to real-life user preferences.

VI. Discussion

x

VII. Conclusion

We deem this project as testable and having a quantifiable outcome, while being under the scope of this course and having manageable workload for a course project.

If we are asked to expand our work, we might simulate a "smart insole" which automatically optimizes stiffness according to sensor readings from the specific user, and if the parameter space has a high dimension, it might require parameter search algorithms.

References

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[2] K. A. McDonald, R. H. Teater, J. P. Cruz, J. T. Kerr, G. Bastas, and K. E. Zelik, “Adding a toe joint to a prosthesis: walking biomechanics, energetics, and preference of individuals with unilateral below-knee limb loss,” *Scientific Reports*, vol. 11, no. 1, p. 1924, 2021.

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